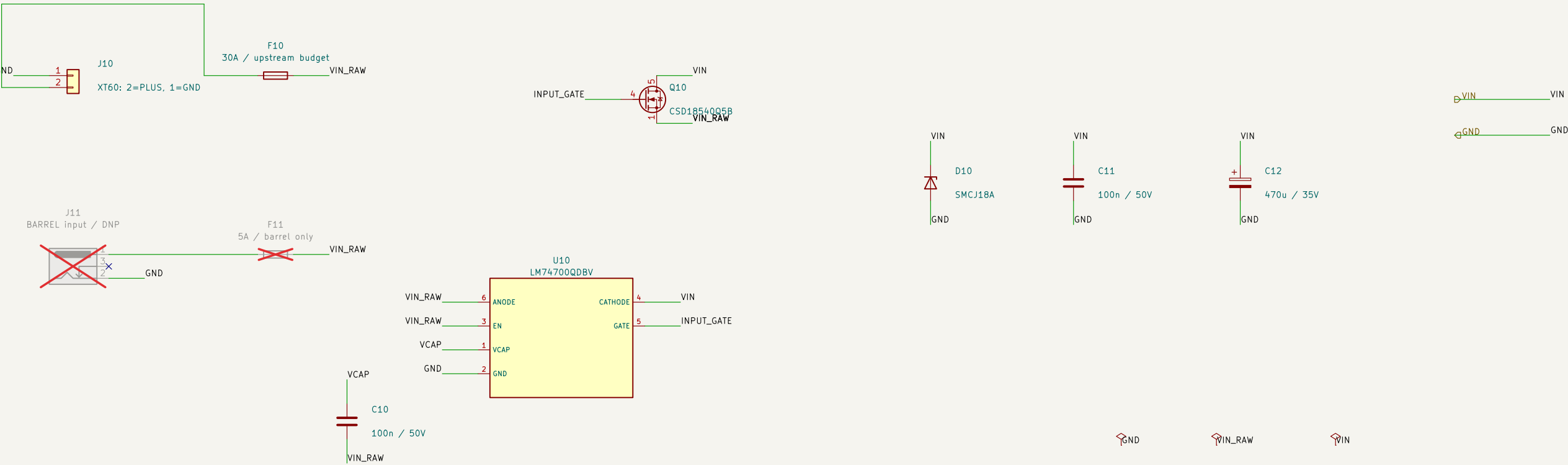
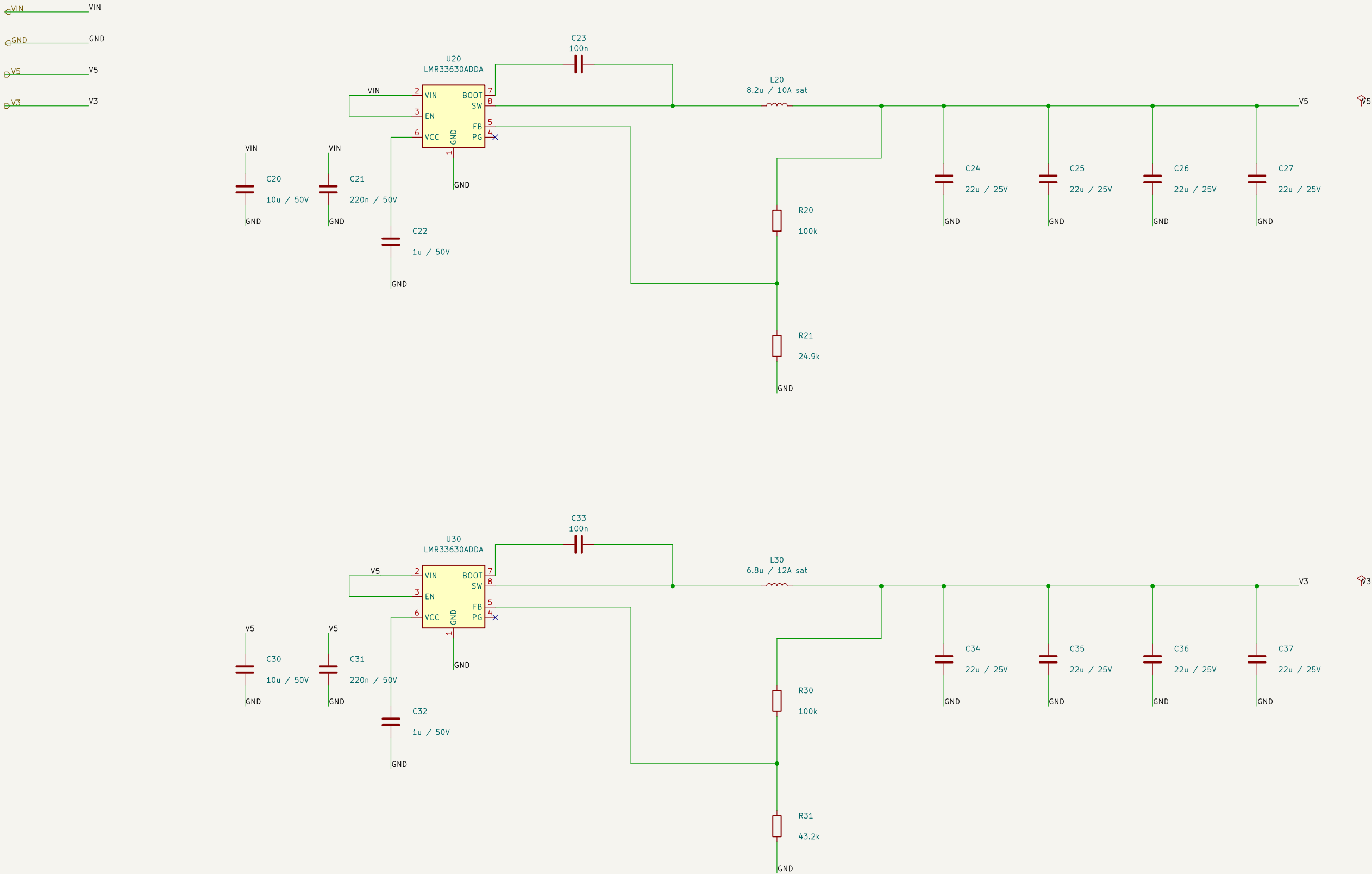


POWER INPUT | 12–18 V nominal | XT60 or barrel assembly option



FIT J10 + F10 OR J11 + F11. Do not populate both input connector paths.
XT60: provisional 25 A shared continuous budget; fuse, copper and temperature still require validation.
Barrel: 5 A absolute TOTAL ceiling; 3.5 A provisional continuous budget after fuse derating. Use a current-limited supply.
LM74700 + Q10 provides reverse-polarity / reverse-current protection. It is not an overvoltage disconnect.
Never apply more than 18 V nominal. TVS pulse capability is not a sustained overvoltage rating.

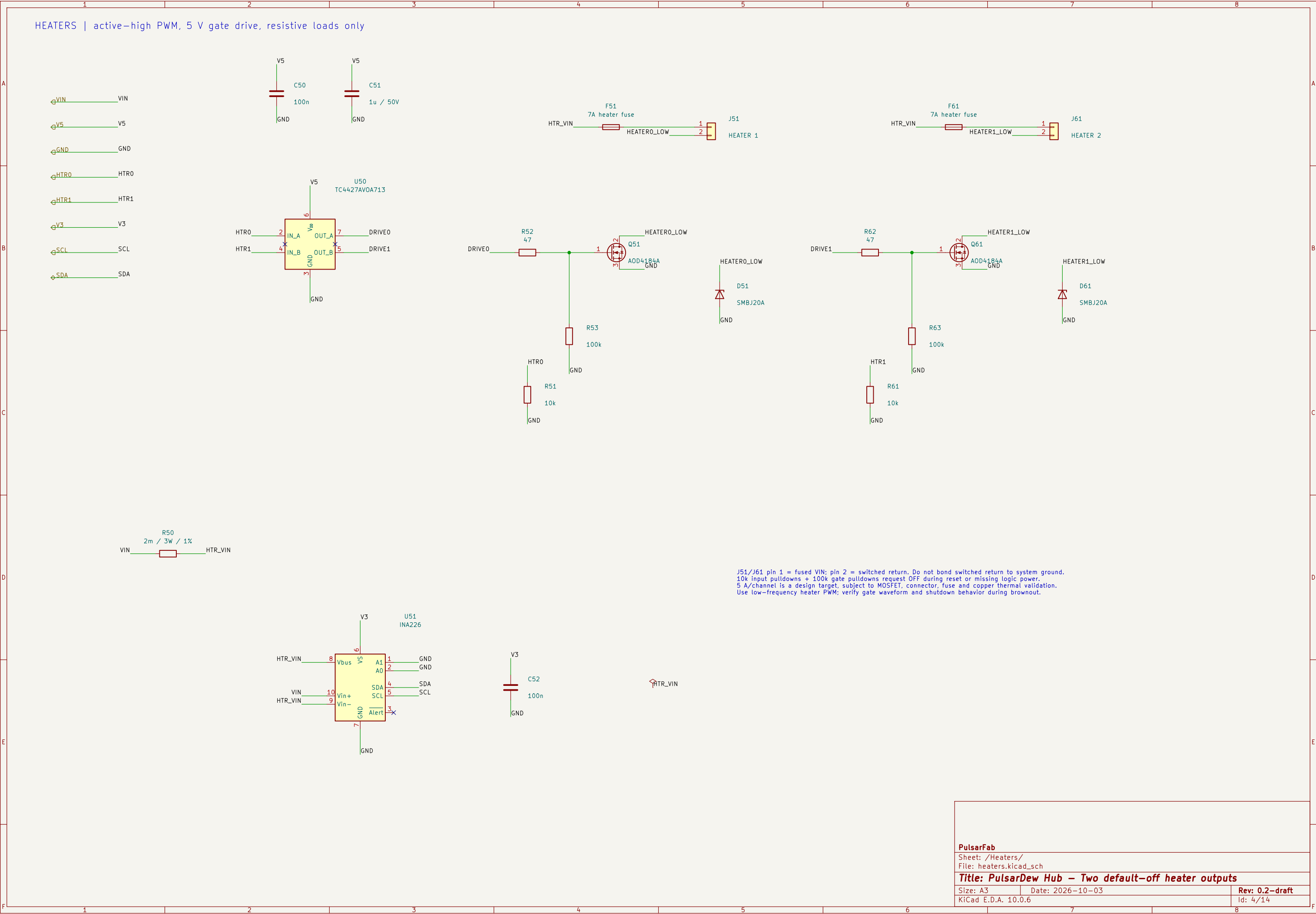
DC SUPPLIES | LMR33630ADDA, 400 kHz | 5 V / 3 A + 3.3 V logic



Follow TI SNVSN3F Table 9–2. 5 V uses nearest preferred 8.2 uH to the 8 uH example.
Place ceramic input/boot/VCC capacitors at the IC; keep SW copper compact and FB away from SW.
USB budget: 4 x 500 mA plus about 0.4 A for logic at full activity. Verify inductor saturation, MLCC bias, transient response and regulator temperature before routing release.

PulsarFab		
Sheet: /DC supplies/ File: supplies.kicad_sch		
Title: PulsarDew Hub – 5 V USB and 3.3 V logic supplies		
Size: A3	Date: 2026–10–03	Rev: 0.2–draft
KiCad E.D.A. 10.0.6		Id: 3/14

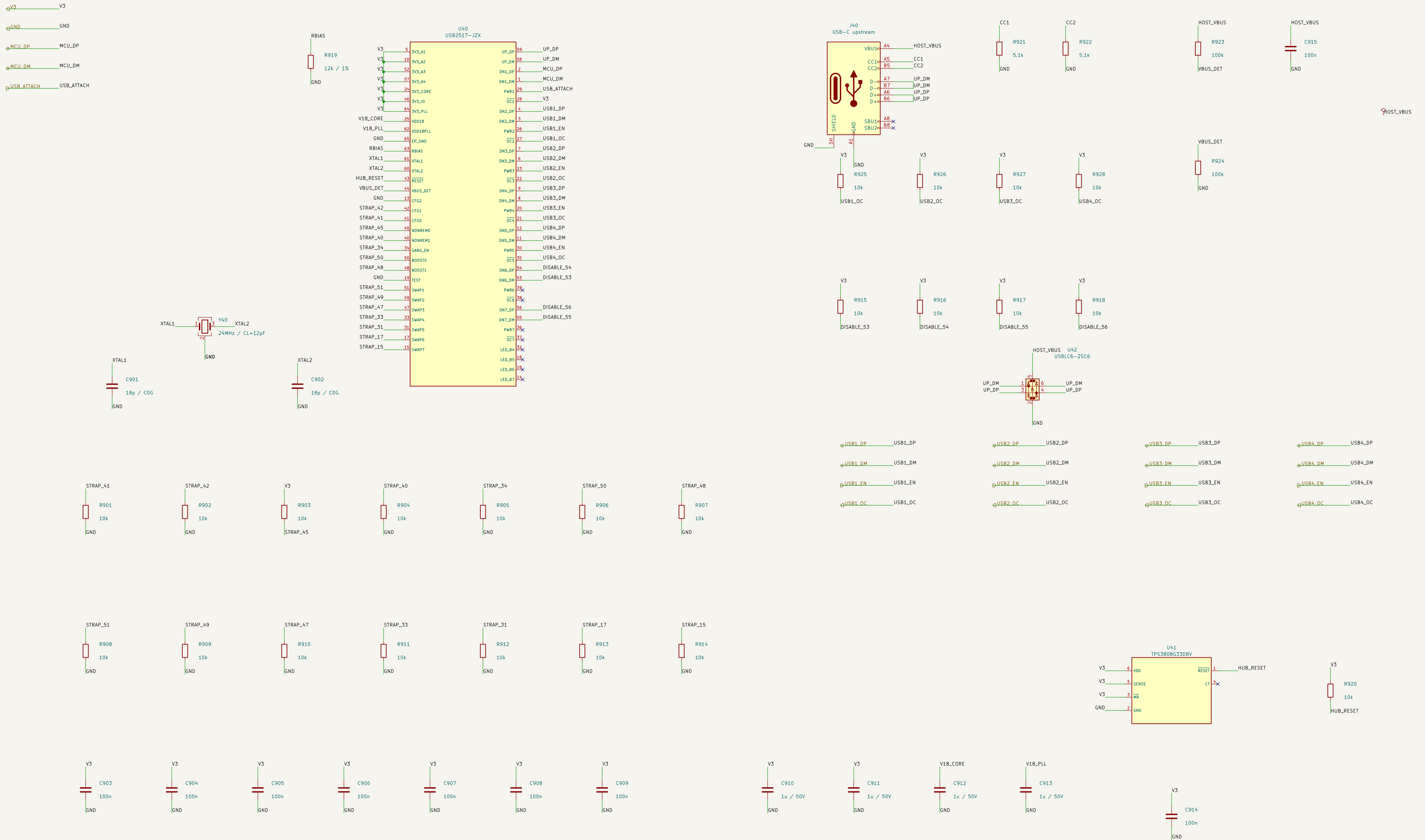
HEATERS | active-high PWM, 5 V gate drive, resistive loads only



[illegible]

Sheet: /Controller/ File: controller.kicad_sch		
Title: PulsarDew Hub – STM32 controller and ambient sensor		
Size: A3	Date: 2026–10–03	Rev: 0.2–draft
KiCad E.D.A. 10.0.6		Id: 5/14

USB HUB | USB2517 | internal port 1 + external ports 2-5; ports 6/7 disabled



Straps: CFG=000 self-powered; NONREM=01 makes port 1 permanent; individual power/OC; normal polarity.
Unused ports 6/7; both D+ and D- pulled to 3V3 through 10k; PWR outputs unused, OC inputs unused.
VDD18 and VDD18PLL are separate INTERNAL regulator outputs; capacitors only; do not connect them together.
Supervisor: 3.07 V threshold, 20 ms delay. Crystal caps start at 18 pf; verify actual board parasitics/drive level.
Keep an uninterrupted ground plane under every USB pair. Verify 90-ohm geometry with the ordered stackup.

SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

VIN VIN
V3 V3
GND GND
ENABLE ENABLE
CURRENT CURRENT
PGOOD PGOOD

VIN VIN
D102 SMBJ18A
GND GND
C101 100n / 50V
GND GND

U101 TPS2598270NRGE
VIN 1 IN OUT 17 OUT
VIN 2 IN OUT 18 OUT
VIN 3 IN OUT 19 OUT
VIN 16 IN OUT 20 OUT
VIN 25 EP_IN OUT 21 OUT
EN_SW 6 EN_UVLO OUT 22 OUT
GND 4 GND OUT 23 OUT
GND 5 GND OUT 24 OUT
GND 14 GND PG PGOOD
GND 26 EP_GND IMON IMON_RAW
IMON 8 ILIM
dVdT 15 DVDT
ITIMER 7 ITIMER
RETRY_DLY 10 RETRY_DLY
NRETRY 11 NRETRY
LDSTRT 12 LDSTRT

OUT OUT
C102 100n / 50V
GND GND

J101 5A DC / center +
OUT OUT
GND GND

V3 V3
R106 10k
PGOOD PGOOD

OUT OUT
D101 SS54
GND GND

VIN VIN
R101 220k UV_SENSE
UV_SENSE UV_SENSE
R102 10k GND
GND GND

VIN VIN
R103 470k OV_SENSE
OV_SENSE OV_SENSE
R104 10k GND
GND GND

U102 TPS3700DDC
V3 5 VDD OUTA 1 EN_SW
UV_SENSE 3 INA+ OUTB 6 EN_SW
OV_SENSE 4 INB-
GND 2 GND

V3 V3
C105 100n
GND GND

ENABLE R110 10k EN_SW
DVDT C103 4.7n / 50V
GND GND
ILIM R107 249 / 1%
GND GND
EN_SW R105 100k
GND GND

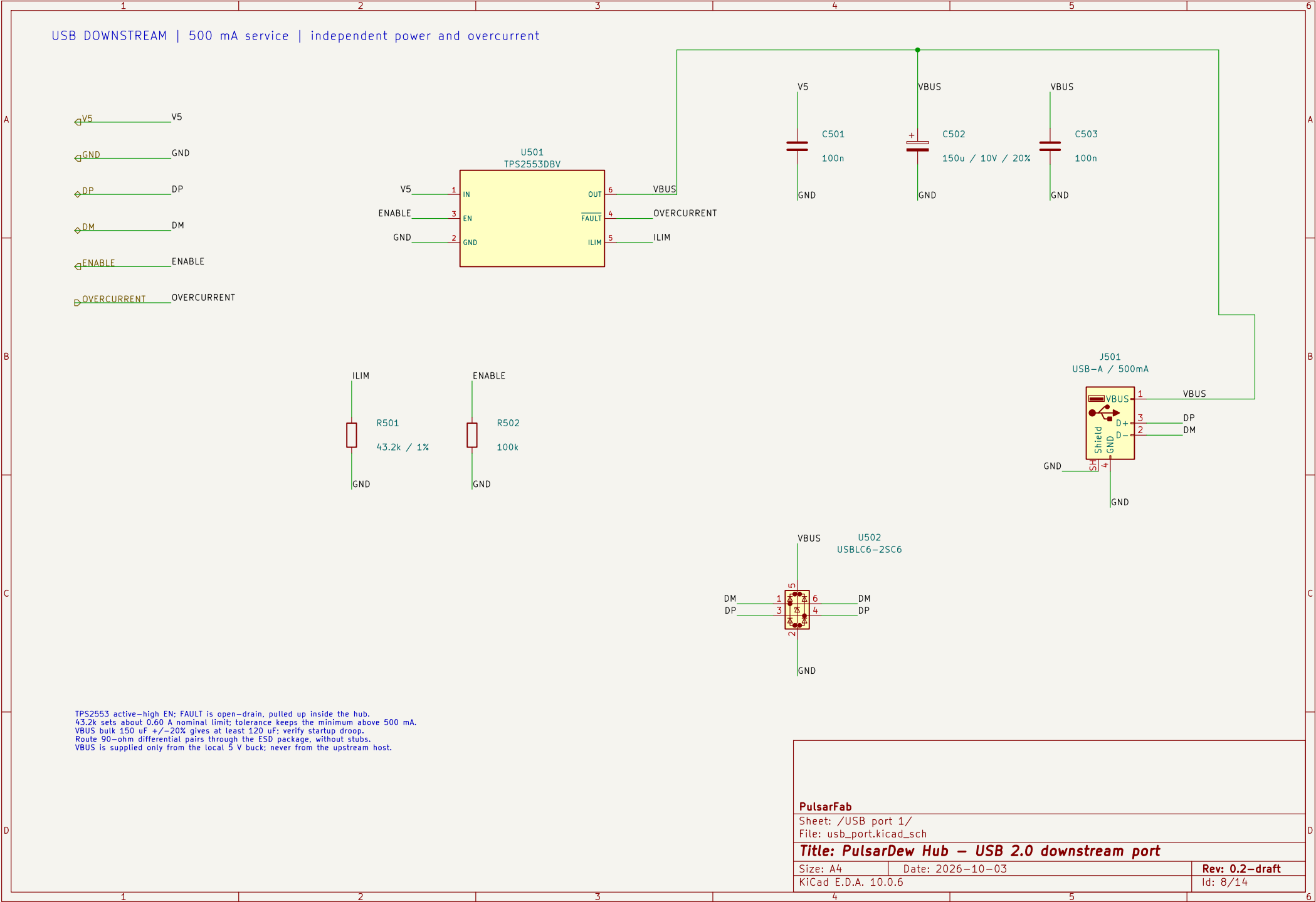
IMON_RAW R109 10k CURRENT
IMON_RAW IMON_RAW
R108 820 / 1%
GND GND
CURRENT C104 10n
GND GND

TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification.
ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off.
IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC filter. Calibrate each outlet.
TPS3700 independently inhibits EN below ~9.2 V or above ~19.2 V; window recovery can re-enable an asserted GPIO.
PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault.
EP25 = VIN, EP26 = GND. -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT.
SMBJ18A input TVS + SS54 output clamp need short loops and hot-plug/inductive-load bench validation.

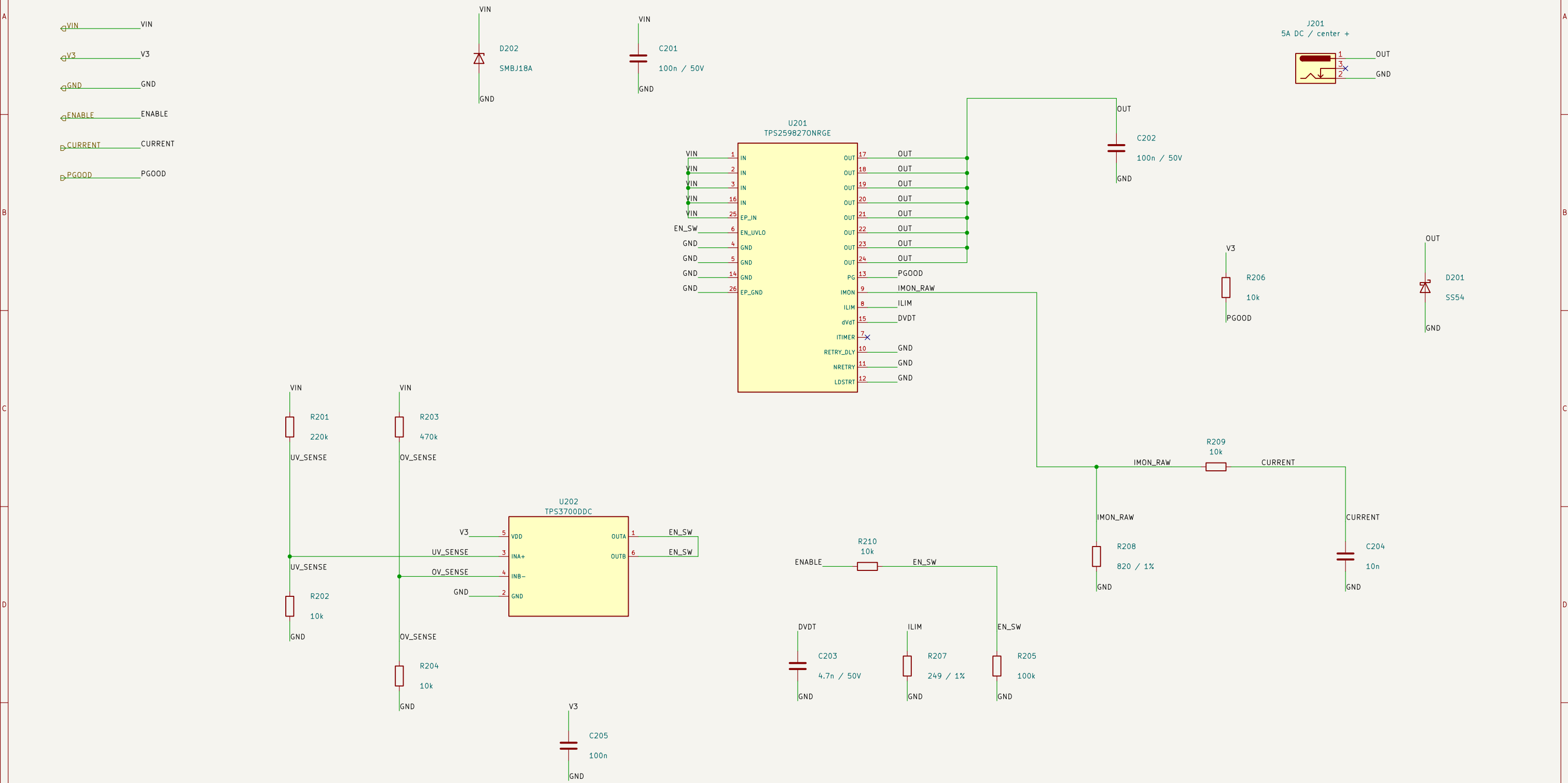
PulsarFab
Sheet: /DC outlet 1/
File: dc_outlet.kicad_sch
Title: PulsarDew Hub - 5 A switched DC outlet
Size: A3 Date: 2026-10-03 Rev: 0.2-draft
KiCad E.D.A. 10.0.6 Id: 7/14

TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification. ITIMER open: fastest overload response. Fast short trip ~2.1s setpoint. RETRY_DLY grounded: latch off. IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V x 5 A); 10k / 10n ADC divider. Calibrate each outlet. TPS37700 independently inhibits EN below -9.2 V or above -19.2 V: window recovery can re-enable an asserted GPIO. PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault. EP25 = VIN, EP26 = GND. -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT. SMBJ18A input TVS + S554 output clamp need short loops and hot+plug/inductive-load bench validation.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

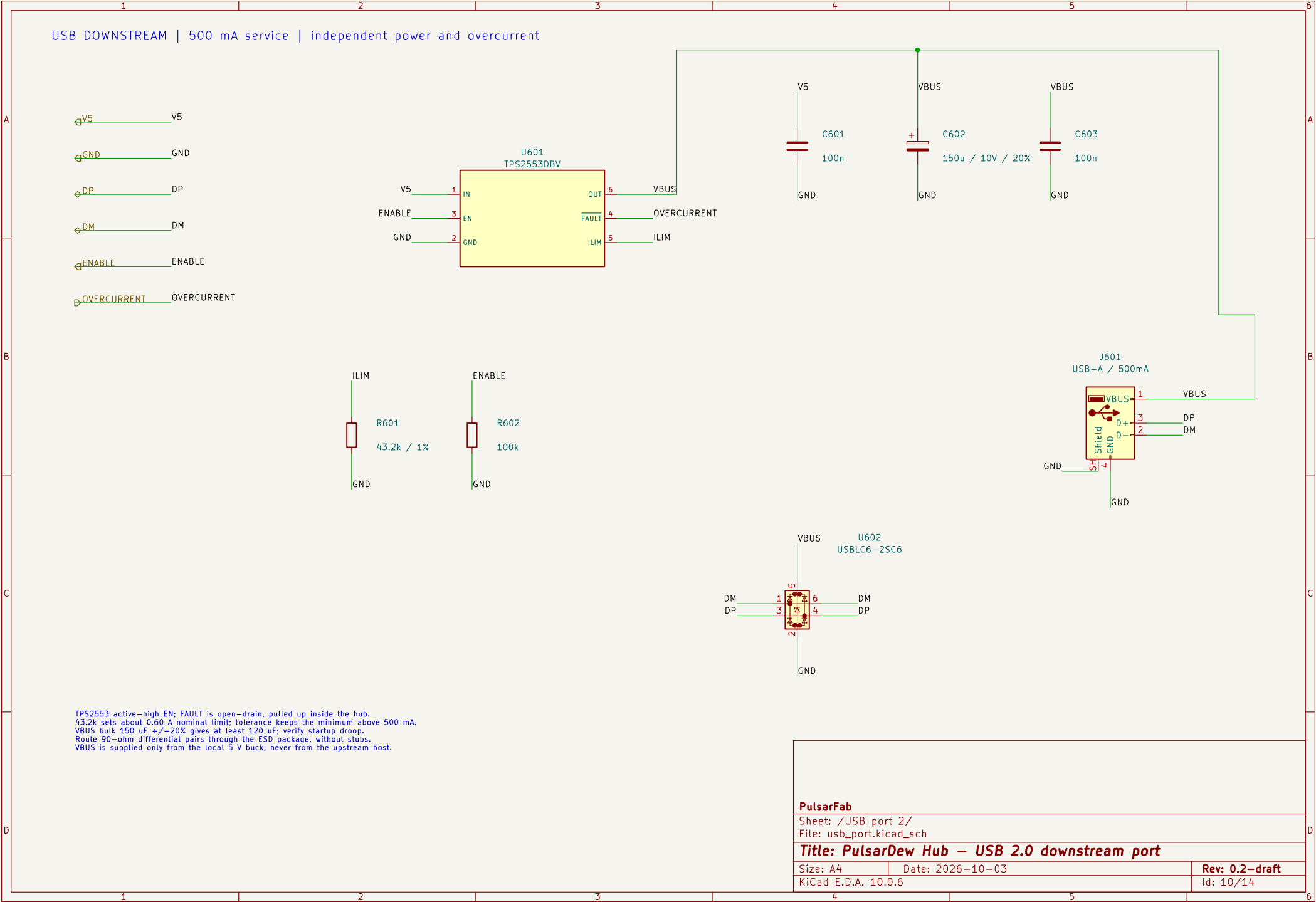


SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker



TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification.
ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off.
IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC filter. Calibrate each outlet.
TPS3700 independently inhibits EN below ~9.2 V or above ~19.2 V; window recovery can re-enable an asserted GPIO.
PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault.
EP25 = VIN, EP26 = GND. ~15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT.
SMBJ18A input TVS + SS54 output clamp need short loops and hot-plug/inductive-load bench validation.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

VIN VIN
V3 V3
GND GND
ENABLE ENABLE
CURRENT CURRENT
PGOOD PGOOD

VIN
D302
SMBJ18A
GND

VIN
C301
100n / 50V
GND

U301
TPS2598270NRGE

VIN 1 IN
VIN 2 IN
VIN 3 IN
VIN 16 IN
VIN 25 EP_IN
EN_SW 6 EN_UVLO
GND 4 GND
GND 5 GND
GND 14 GND
GND 26 EP_GND

OUT 17 OUT
OUT 18 OUT
OUT 19 OUT
OUT 20 OUT
OUT 21 OUT
OUT 22 OUT
OUT 23 OUT
OUT 24 OUT
PG 13 PGOOD
IMON 9 IMON_RAW
ILIM 8 ILIM
dVdT 15 DVDT
ITIMER 7
RETRY_DLY 10 GND
NRETRY 11 GND
LDSTRT 12 GND

OUT
C302
100n / 50V
GND

V3
R306
10k
PGOOD

OUT
D301
SS54
GND

VIN
R301
220k
UV_SENSE

VIN
R303
470k
OV_SENSE

V3 5 VDD
UV_SENSE 3 INA+
OV_SENSE 4 INB-
GND 2 GND

U302
TPS3700DDC

OUTA 1 EN_SW
OUTB 6 EN_SW

UV_SENSE
R302
10k
GND

OV_SENSE
R304
10k
GND

V3
C305
100n
GND

ENABLE
R310
10k
EN_SW

DVDT
C303
4.7n / 50V
GND

ILIM
R307
249 / 1%
GND

EN_SW
R305
100k
GND

IMON_RAW
R308
820 / 1%
GND

IMON_RAW
R309
10k
CURRENT

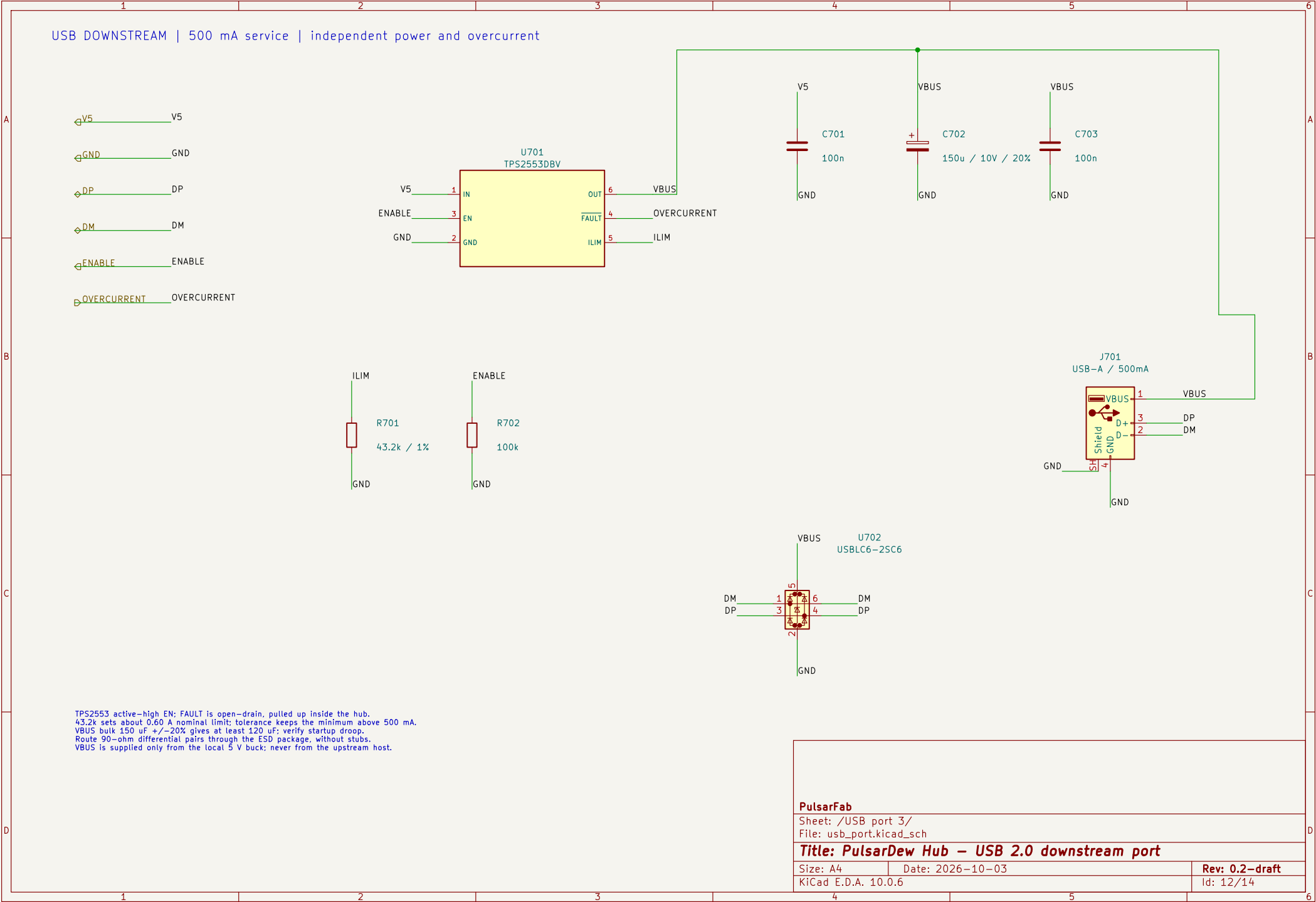
CURRENT
C304
10n
GND

TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification.
ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off.
IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC filter. Calibrate each outlet.
TPS3700 independently inhibits EN below ~9.2 V or above ~19.2 V; window recovery can re-enable an asserted GPIO.
PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault.
EP25 = VIN, EP26 = GND. -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT.
SMBJ18A input TVS + SS54 output clamp need short loops and hot-plug/inductive-load bench validation.

PulsarFab
Sheet: /DC outlet 3/
File: dc_outlet.kicad_sch
Title: PulsarDew Hub – 5 A switched DC outlet
Size: A3 Date: 2026–10–03 Rev: 0.2–draft
KiCad E.D.A. 10.0.6 Id: 11/14

TPS2598270: 249R gives 5.9/ A nominal breaker threshold; 5 A service target needs tolerance/bench qualification. ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_OLY grounded: latch off. IMON: 246 uA x 620R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC divider. Calibrate each outlet. TPS37700 independently inhibits EN below -9.2 V or above -19.2 V; window recovery can re-enable an asserted GPIO. PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault. EP26 = VIN, EP26 = GND, -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT. SMBJ18A input TVS + 555A output clamp need short loops and hot+plug/inductive-load bench validation.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



PulsarFab

Sheet: /USB port 3/
File: usb_port.kicad_sch

Title: PulsarDew Hub – USB 2.0 downstream port

Size: A4

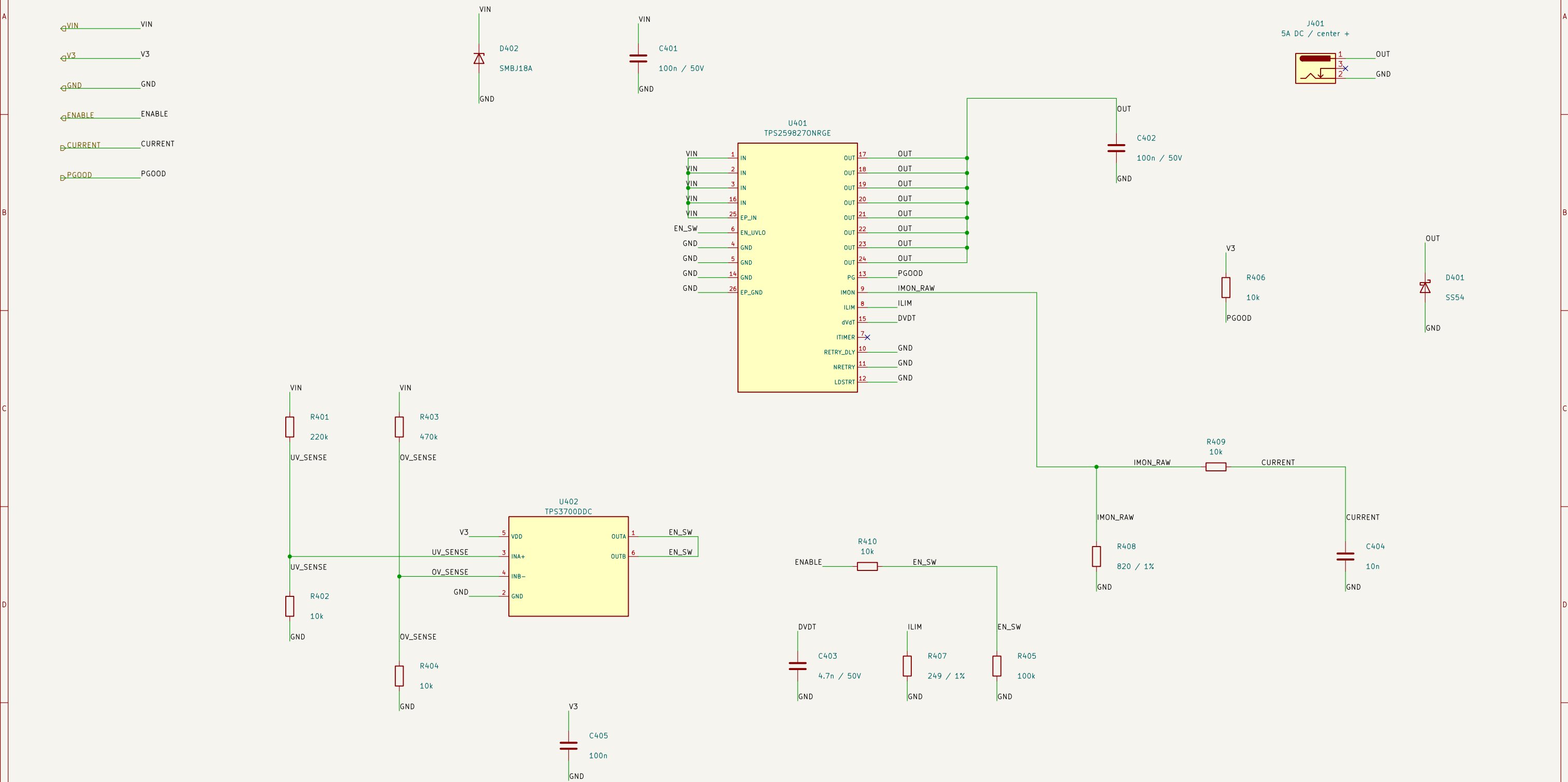
Date: 2026-10-03

Rev: 0.2-draft

KiCad E.D.A. 10.0.6

Id: 12/14

SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker



TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification.
ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off.
IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC filter. Calibrate each outlet.
TPS3700 independently inhibits EN below ~9.2 V or above ~19.2 V; window recovery can re-enable an asserted GPIO.
PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault.
EP25 = VIN, EP26 = GND. ~15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT.
SMBJ18A input TVS + SS54 output clamp need short loops and hot-plug/inductive-load bench validation.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

